

CO2 - ASSESSMENT 1

DSA-0202 Computer Vision With Open CV

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SLOT-D

Question 1: Salt-and-Pepper Noise Removal Before Edge Detection

a. Most suitable filtering technique

Median Filter (3×3 or 5×5)

- Best filter for salt-and-pepper (impulse) noise.
- Removes isolated black and white pixels.
- Preserves edges much better than linear filters.

b. Why is a median filter better than an averaging filter?

Median Filter	Averaging Filter
Removes impulse noise effectively	Spreads noisy pixels to neighboring pixels
Preserves edges	Blurs edges
Uses median value	Uses mean value
Robust against outliers	Highly affected by noisy pixels

Example

Pixel values:

20 22 21

23 255 24

22 21 23

- Average = 48 (incorrect)
- Median = 22 (correct)

Hence, the median filter restores the original intensity better.

c. Probability that a noisy center pixel will be corrected

Given:

- Noise density = **0.05**
- Window = **3×3**
- Total pixels in window = **9**

Assumption:

The center pixel is noisy while most neighboring pixels are not.

Probability that a neighboring pixel is not noisy

$$P = 1 - 0.05 = 0.95$$

Expected clean neighbors

$$8 \times 0.95 = 7.6$$

Since most neighboring pixels are clean, the median will almost always be a correct value.

Estimated probability

$$\boxed{\approx 95\% - 99\%}$$

A common approximation:

$$\boxed{P \approx 0.95}$$

d. Total noisy pixels

Image size

$$1024 \times 1024 = 1,048,576$$

Noise density

$$5\% = 0.05$$

Noisy pixels

$$1,048,576 \times 0.05 = 52,428.8$$

Approximately

$$\boxed{52,429 \text{ noisy pixels}}$$

e. Expected noisy pixels remaining

Median filtering removes most impulse noise.

Assume **95%** removed.

Remaining

$$52,429 \times 0.05 = 2,621$$

Approximately

2,600 noisy pixels remain

f. If noise is Gaussian instead

Median filter is no longer the best choice.

Suitable filters:

- Gaussian Filter
- Wiener Filter
- Bilateral Filter

Reason:

Gaussian noise affects every pixel slightly.

Median filtering is designed mainly for impulse noise.

Question 2: Discretization of a Continuous Image

a. Conversion into digital image

Two steps are required:

Sampling

- Converts continuous coordinates into pixels.
- Produces discrete x and y positions.

Quantization

- Converts continuous intensity into fixed gray levels.

Thus,

Continuous Image

↓

Sampling

↓

Quantization

↓

Digital Image

b. Spatial variation

The image intensity follows a **sinusoidal pattern** along the x-direction.

So,

$$I(x) = A\sin(2\pi fx)$$

Along y-direction:

Intensity remains constant.

c. Minimum sampling requirement

According to the **Nyquist Sampling Theorem**,

Sampling frequency

$$f_s \geq 2f$$

where

- f =highest spatial frequency

Therefore,

Minimum sampling rate

$$\boxed{2f}$$

d. What happens below sampling rate?

Aliasing occurs.

Effects:

- Wrong patterns appear.
- False frequencies appear.
- Jagged edges.
- Information loss.

e. Suitable quantization

Use

8-bit grayscale

Number of levels

$$2^8 = 256$$

Intensity values

$$0 - 255$$

This is the standard grayscale representation.

f. Effect of quantization

Image Quality

Higher bits

→ Better quality

Lower bits

→ Visible banding

Storage

8-bit image

Requires

1 byte/pixel

Intensity Accuracy

Higher bit depth

→ Better intensity precision

Question 3: Edge Detection in Thin Line Images

a. Comparison

Feature	Sobel	Prewitt	Canny
Accuracy	Good	Moderate	Excellent
Noise handling	Moderate	Poor	Excellent
Thin edge detection	Moderate	Moderate	Best
Edge localization	Good	Good	Excellent
Real applications	Medium	Medium	Best

Best choice: Canny

b. Sobel gradient response

Standard Sobel operators

Horizontal

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

Vertical

$$G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

For a horizontal edge,

$$G_x \approx 0$$

$$G_y = \text{Maximum}$$

Gradient magnitude

$$G = \sqrt{G_x^2 + G_y^2}$$

Since

$$G_x = 0$$

then

$$G = |G_y|$$

c. Gradient magnitude and direction

Gradient Magnitude

- Shows edge strength.
- Larger value $\rightarrow$ stronger edge.

Gradient Direction

Shows orientation.

Examples:

- $0^\circ \rightarrow$ Vertical edge
- $90^\circ \rightarrow$ Horizontal edge
- $45^\circ \rightarrow$ Diagonal edge
-

d. Role of Gaussian smoothing

Before edge detection,

Gaussian filter removes

- Noise

- Small intensity fluctuations

Result

- Fewer false edges
- Better edge detection

e. Effect on thin lines

Too much Gaussian smoothing

- Blurs thin lines
- Reduces edge sharpness
- Slightly shifts edge location

Therefore,

Small Gaussian kernels are preferred.

f. Threshold reduced by half

Expected effects:

- More edges detected
- More weak edges retained
- More false edges
- Increased noise detection

Question 4: Feature Extraction for Handwritten Digit Recognition

a. Working principles

SIFT

- Detects scale-space extrema.
- Computes orientation.
- Creates 128-dimensional descriptor.

SURF

- Uses Hessian matrix.
- Faster than SIFT.
- Uses integral images.

ORB

- Uses FAST keypoint detector.
- Uses BRIEF descriptor.

- Very fast.

b. Comparison

Feature	SIFT	SURF	ORB
Scale invariant	Yes	Yes	Partial
Rotation invariant	Yes	Yes	Yes
Speed	Slow	Medium	Fast
Real-time	No	Moderate	Yes

c. Rotated by 45°

Reliable methods:

- SIFT
- SURF
- ORB

SIFT and SURF are generally more robust than ORB.

d. Total keypoints

Average keypoints : 120

Images: 50

Total : $120 \times 50 = 6000$

Answer: 6000 keypoints

e. Suitable dimensionality reduction

Principal Component Analysis (PCA)

Retain : 95% variance

f. Benefits

PCA

- Removes redundant features
- Reduces computation
- Faster training
- Less memory
- Improves classifier performance

Question 5: Morphological Preprocessing for Character Recognition

a. Remove white noise dots

Use

Opening

Opening

= Erosion + Dilation

Removes small white objects.

b. Fill small gaps

Use

Closing

Closing

= Dilation + Erosion

Fills

- Small holes
- Small gaps

c. Difference

Opening	Closing
Removes small white noise	Fills small holes
Smooths outer boundary	Smooths inner boundary
Erosion $\rightarrow$ Dilation	Dilation $\rightarrow$ Erosion

d. Effect of 3×3 structuring element

- Removes tiny noise
- Fills tiny gaps
- Slightly thins or thickens strokes
- Usually preserves character shape

e. Risk of large structuring element

Large structuring element may:

- Remove thin character strokes
- Merge nearby characters
- Distort character shapes

- Reduce recognition accuracy

f. Complete preprocessing pipeline

Image Acquisition



Grayscale Conversion



Noise Removal (Median Filter)



Thresholding (Binary Image)



Morphological Opening



Morphological Closing



Character Segmentation



Normalization (Resize)



Feature Extraction (SIFT / ORB / HOG)



Classification (SVM / CNN / KNN)

This pipeline removes noise, enhances characters, extracts meaningful features, and prepares the data for accurate handwritten character recognition.